Fundamentals-of-AI-and-ML-Evaluated-Course-Project

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Bring Your Own Project (BYOP)

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INTRODUCTION

In academic life, time management is considered to be one of the greatest challenges for students. Students may think that they are studying hard, but without tracking, it is difficult to evaluate their productivity.

The Student Productivity Tracker is a command-line-based Python program developed with the aim of tracking student productivity. This program is designed in such a way that it can track student progress.

This project is built with ease and simplicity, and it uses basic programming concepts.

PROBLEM STATEMENT

The students are facing some problems in their productivity, which are as follows:
Lack of knowledge about the actual study time
Irregular pattern of study
There is no way to track their study
Difficulty in achieving their study goals

Students are making assumptions about their study, which leads to inefficiency in their work.

This project can solve the above-mentioned problem by creating a system for tracking students' study hours.

OBJECTIVES

- The main objectives of this project are as follows:
- To design a CLI-based application using Python
- To allow students to log their study hours
- To implement data storage using file handling
- To improve students' understanding of Python concepts

METHODOLOGY

The development methodology followed in creating the project is as follows:

Problem Identification

The problem of poor productivity tracking among students was identified.

Solution Planning

The solution planning was carried out by choosing a command-line tool that would allow for focus on logic rather than interface complexity.

System Design

The system design was carried out by breaking it down into modules that include input handling, data storage, data processing, and output display.

Implementation

The implementation was carried out by creating Python functions for all modules.

Testing

The system was tested with inputs to ensure correctness and stability.

TOOLS AND TECHNOLOGIES USED

1. Programming Language: Python
2. Concepts Used: File Handling (open, read, write)
3. Functions
4. Loops
5. Conditional Statements
6. Exception Handling
7. Execution Environment: Command Line Interface

No external libraries or modules were used.

SYSTEM DESIGN

The system is menu-driven and interactive. The components of the system include:

Input Module – study hours, goals
Storage Module – uses text files for storing data
Processing Module – performs calculations
Output Module – uses terminal for displaying results

IMPLEMENTATION DETAILS

Adding Study Hours

The user is prompted to enter their study hours, which are saved in a file.

Viewing Total Hours

The program reads the saved data and calculates the total hours.

Viewing History

The program displays all entries in sequence.

Setting Weekly Goals

The user is prompted to enter their goal, which is saved in another file.

Tracking Progress

The program compares the total hours with the user's goal.

Reset Function

The user can reset all saved data if required.

CHALLENGES FACED

- Managing file handling without extra modules
- Dealing with invalid data input
- Data persistence
- Simple menu management

All these challenges were overcome with testing and debugging.

RESULTS AND OUTPUT

- The program is successful in:
- Tracking study hours
- Data persistence
- Showing user progress
- Helping users develop disciplined habits

ADVANTAGES

1. Simple program with easy user interface
2. No extra dependencies required
3. Works on any computer with Python installed
4. Helps users develop disciplined habits

LIMITATIONS

1. No graphical user interface
2. Data visualization is limited
3. Single-user program

FUTURE ENHANCEMENTS

- Data can be tracked daily
- Data can be visualized
- Multi-user support can be added
- The program can be run online

CONCLUSION

This project has shown that it is possible to transform a simple idea into something practical and applicable through the use of Python. This project also reinforces core programming concepts in solving a real-world problem.

Student Productivity Tracker is a great starting point for developing sophisticated applications in the future.